

TEAM

THE GRAY LAMP

(المصباح الرمادي)

PRESENTS

Smart Parking Management System

MEMBERS:

- MOHAMED MESBAH
- MOSTAFA ELRAMADY



PROBLEMS:

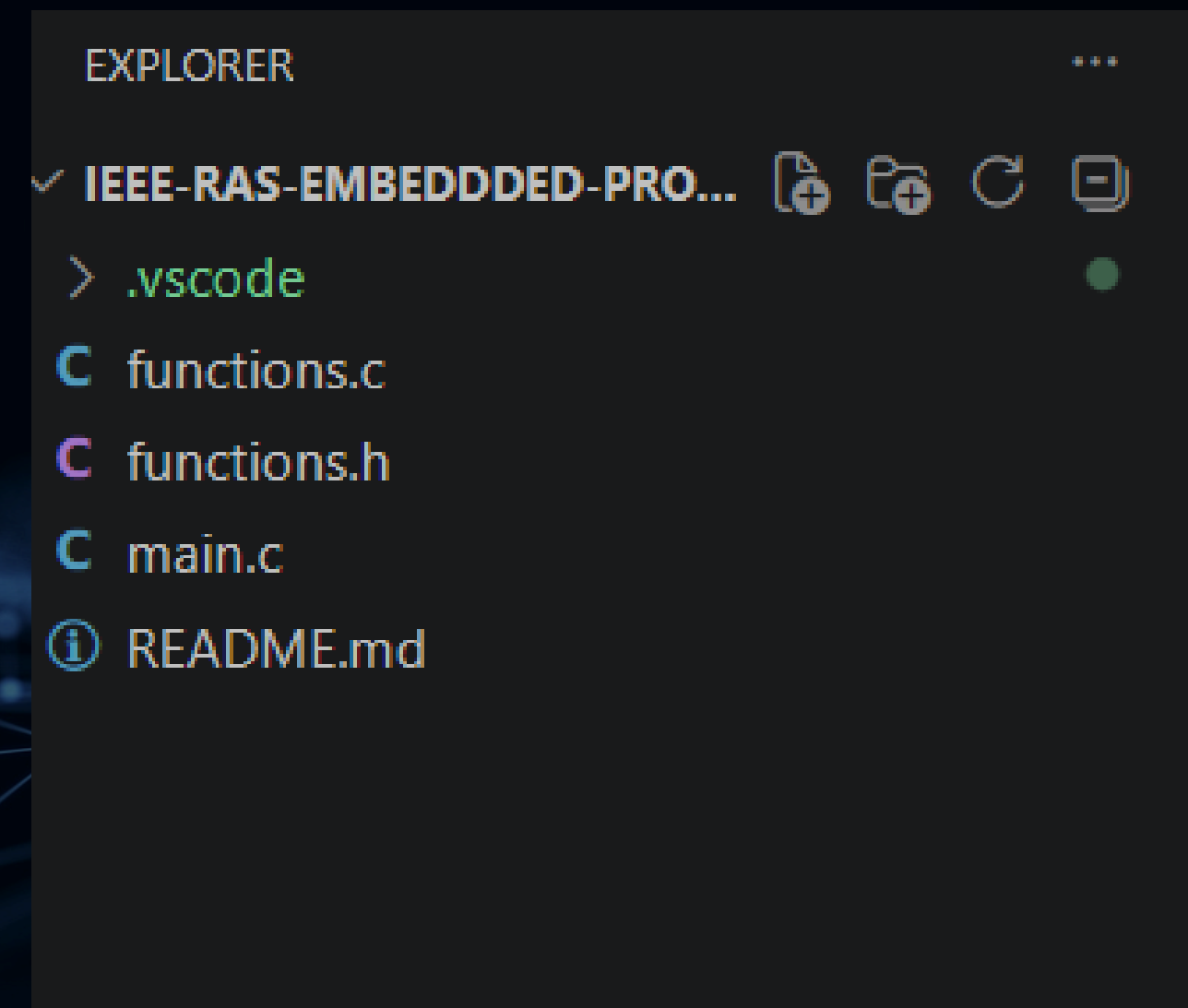
- **TRADITIONAL MANAGEMENT: MANUAL PARKING ALLOCATION CAUSES SEVERE DELAYS AND TRAFFIC LOG CONFLICTS**
- **NO REAL-TIME TRACKING: HARD TO TRACK EXACT VEHICLE CHECK-IN DURATION AND DYNAMIC BILLINGS.**

OBJECTIVE:

- **AUTOMATED SYSTEM: BUILD A LIGHTWEIGHT, HIGH-PERFORMANCE C APPLICATION FOR INSTANT SLOT ASSIGNMENT.**
- **DATA SECURITY: ENSURE ALL SYSTEM LOGS ARE DYNAMICALLY SECURED AGAINST POWER OR SYSTEM FAILURE.**

SOFTWARE ARCHITECTURE

- **MAIN.C:** HANDLES CORE APPLICATION EXECUTION LOOP AND INTERACTIVE MENUS.
- **FUNCTIONS.C:** IMPLEMENTS INTERNAL ALLOCATION ALGORITHMS, SORTING, AND FILE I/O OPERATIONS.
- **FUNCTIONS.H:** DEFINES SHARED STRUCTURES, GLOBAL MACROS, AND FUNCTION PROTOTYPES.



SYSTEM OUTPUT & DASHBOARD

- CLEAN TERMINAL-BASED INTERACTIVE UI.
- SECURE CHOICES FOR CHECK-IN, CHECK-OUT, AND LOGS.

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

    === SMART PARKING MANAGEMENT SYSTEM ===
    1. Park a New Vehicle
    2. Vehicle Exit & Checkout (Leave)
    3. View Parking Overview & Available Slots
    4. Search for a Vehicle
    5. Edit Vehicle Information
    6. Exit Program
    =====
    Enter Your Choice: █
```


ADDING VEHICLES

```
C:\Users\HP\Desktop\ × + v
=== SMART PARKING MANAGEMENT SYSTEM ===
1. Park a New Vehicle
2. Vehicle Exit & Checkout (Leave)
3. View Parking Overview & Available Slots
4. Search for a Vehicle
5. Edit Vehicle Information
6. Exit Program
=====
Enter Your Choice: 1

Enter Vehicle ID: 10
Enter Owner Name: mesbah
Enter Plate Number: mes77
Enter Vehicle Type: malaky

[Success] Vehicle record created successfully.

[Success] Slot 1 assigned to Vehicle ID 10.

[Success] Data saved to parking.txt successfully.

Press [Enter] key to continue...
```

```
=== SMART PARKING MANAGEMENT SYSTEM ===
1. Park a New Vehicle
2. Vehicle Exit & Checkout (Leave)
3. View Parking Overview & Available Slots
4. Search for a Vehicle
5. Edit Vehicle Information
6. Exit Program
=====
Enter Your Choice: 1

Enter Vehicle ID: 20
Enter Owner Name: mostafa
Enter Plate Number: mo66
Enter Vehicle Type: malaky

[Success] Vehicle record created successfully.

[Success] Slot 2 assigned to Vehicle ID 20.

[Success] Data saved to parking.txt successfully.

Press [Enter] key to continue...|
```

DATA PERSISTENCE

- DATA IS PERMANENTLY SAVED TO LOCAL FILES (.TXT / .DAT).
- PREVENTS DATA LOSS UPON SYSTEM SHUTDOWN.

```
File Edit View
/2
10
mesbah
mes77
maLaky
1783202774
1
20
mostafa
mo66
maLaky
1783202790
2
```

SYSTEM TESTING & EDGE CASES

BULLET POINTS:

- **SLOT CONFLICT PREVENTION:** SYSTEM IMMEDIATELY REJECTS AND BLOCKS ASSIGNING A VEHICLE TO AN ALREADY OCCUPIED SLOT.
- **POST-DELETION VERIFICATION:** ATTEMPTING TO VIEW OR ACCESS A REMOVED VEHICLE'S DATA GUARANTEES AN IMMEDIATE "NOT FOUND" VALIDATION MESSAGE.
- **INVALID ID TRACKING:** MODIFYING VEHICLE RECORDS USING A NON-EXISTING ID IS FULLY TRAPPED AND BLOCKED BY INPUT FILTERS.

```
C:\Users\HP\Desktop\ x + v

=== SMART PARKING MANAGEMENT SYSTEM ===
1. Park a New Vehicle
2. Vehicle Exit & Checkout (Leave)
3. View Parking Overview & Available Slots
4. Search for a Vehicle
5. Edit Vehicle Information
6. Exit Program
=====
Enter Your Choice: 1

Enter Vehicle ID: 10
[Error] Vehicle ID 10 already exists in the system!

[Success] Data saved to parking.txt successfully.

Press [Enter] key to continue...|
```


SEARCH FOR A VEHICLE

BY PLATE NUMBER

```
C:\Users\HP\Desktop\ x + v

=== SMART PARKING MANAGEMENT SYSTEM ===
1. Park a New Vehicle
2. Vehicle Exit & Checkout (Leave)
3. View Parking Overview & Available Slots
4. Search for a Vehicle
5. Edit Vehicle Information
6. Exit Program
=====
Enter Your Choice: 4

--- Search Menu ---
1. Search By Vehicle ID
2. Search By Plate Number
Enter Choice: 2

Enter Plate Number to search: mes77

--- Vehicle Details ---
Vehicle ID   : 10
Owner Name   : mesbah
Plate Number : mes77
Vehicle Type : malaky
Slot Number  : Slot 1
Entry Time   : Sun Jul 05 01:06:14 2026
-----

Press [Enter] key to continue...
```

BY ID

```
C:\Users\HP\Desktop\ x + v

=== SMART PARKING MANAGEMENT SYSTEM ===
1. Park a New Vehicle
2. Vehicle Exit & Checkout (Leave)
3. View Parking Overview & Available Slots
4. Search for a Vehicle
5. Edit Vehicle Information
6. Exit Program
=====
Enter Your Choice: 4

--- Search Menu ---
1. Search By Vehicle ID
2. Search By Plate Number
Enter Choice: 1

Enter Vehicle ID to search: 10

--- Vehicle Details ---
Vehicle ID   : 10
Owner Name   : mesbah
Plate Number : mes77
Vehicle Type : malaky
Slot Number  : Slot 1
Entry Time   : Sun Jul 05 01:06:14 2026
-----

Press [Enter] key to continue...|
```

VIEW PARKING OVERVIEW

```
C:\Users\HP\Desktop\ ^ T ^
=== SMART PARKING MANAGEMENT SYSTEM ===
1. Park a New Vehicle
2. Vehicle Exit & Checkout (Leave)
3. View Parking Overview & Available Slots
4. Search for a Vehicle
5. Edit Vehicle Information
6. Exit Program
=====
Enter Your Choice: 3
      === PARKING SYSTEM OVERVIEW ===
Total Parking Slots      : 20
Occupied Slots Real      : 2
Available Slots Real     : 18
Occupancy Percentage     : 10.00%
Total Registered Logs    : 2/100
=====

Press [Enter] key to continue...|
```


NEXT STEPS: HARDWARE INTEGRATION

- PORTING VERIFIED PURE C CORE LOGIC ONTO DEDICATED PHYSICAL MICROCONTROLLERS (ESP32).
 - UTILIZING IR SENSORS FOR LIVE CAR ARRIVAL TRIGGERS AND SERVO MOTORS FOR SMART GATES.
 - LEVERAGING ESP-NOW PROTOCOL LAYER FOR ULTRA-LOW LATENCY WIRELESS NODE COMMUNICATION.
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LARANA, INC.

THANK YOU